A BOUNDED-ABOVE FINITELY PROJECTIVE COMPLEX OVER A HENSELIAN QUOTIENT LIFTS TO THE RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span contains a four-item enumerated hypothesis list; it is delivered verbatim including the phrase 'Moreover, if E^i is free, then P^i is free.' The five references inside the counted proof span (including the Nakayama reference) are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The abstract lifting lemma for complexes with respect to a class of modules closed under the five listed properties. invoked at proof lines 19671-19674 and 19682-19683. Is the engine that produces the complex P^bullet.
- Lifting of finite projective modules across a henselian pair. invoked at proof lines 19677-19679. Supplies property (4) of the lifting lemma.
- A pseudo-coherent complex is represented by a bounded above complex of finite free modules. invoked at proof lines 19680-19681. Supplies property (5) of the lifting lemma.
- Nakayama's lemma and the classification of isomorphism classes of finite projective lifts. invoked at proof lines 19675-19676 and 19683-19684. Gives property (3) and the final freeness assertion.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 145: sources/stacks-more-algebra/pinned-original-source.tex, source lines 145–145. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19390: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19390–19390. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 3348: sources/stacks-more-algebra/pinned-original-source.tex, source lines 3348–3348. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let $E^\bullet$ be a complex of R/I-modules. Let K be an object of $D(R)$. Assume that*

(1) *$E^\bullet$ is a bounded above complex of finite projective R/I-modules,*
(2) *$K \otimes^{\mathbf{L}}_R R/I$ is represented by $E^\bullet$ in $D(R/I)$,*
(3) *K is pseudo-coherent, and*
(4) *(R, I) is a henselian pair.*

Then there exists a bounded above complex $P^\bullet$ of finite projective R-modules representing K in $D(R)$ such that $P^\bullet \otimes_R R/I$ is isomorphic to $E^\bullet$. Moreover, if E^i is free, then P^i is free.

Proof. We apply Lemma 19390 using the class $\mathcal{P}$ of all finite projective R-modules. Properties (1) and (2) of the lemma are immediate. Property (3) follows from Nakayama's lemma (Algebra, Lemma 00DV). Property (4) follows from the fact that we can lift finite projective R/I-modules to finite projective R-modules, see Lemma 3348. Property (5) holds because a pseudo-coherent complex can be represented by a bounded above complex of finite free R-modules. Thus Lemma 19390 applies and we find $P^\bullet$ as desired. The final assertion of the lemma follows from Lemma 145. □